Foundations of Machine Learning and AI

Homework 1 | Fall 2026

1. Betting and expected utility

Let $p = 0.85$ be Victor's subjective probability and $q = 0.60$ the market-implied probability. Assume no fees, no borrowing, and an amount at risk $a \in [0, d]$. Interpreting the market quote as a contract costing q dollars and paying one dollar if Biden wins, the net profit per dollar staked is $b = (1 - q)/q = 2/3$. Final wealth is therefore

$$W(a) = \begin{cases} d + ba, & \text{Biden wins,} \\ d - a, & \text{Biden loses.} \end{cases}$$

Wording ambiguity. Literal "even-money" odds mean $b = 1$, which conflicts with $q = 0.60$. The main solution uses the quoted probability; the literal even-money alternative is given below.

(a) Linear utility

For $U(W) = W$,

$$\mathbb{E}[U(W(a))] = p(d + ba) + (1 - p)(d - a) = d + a\left(\frac{p}{q} - 1\right) = d + \frac{5}{12}a.$$

This is strictly increasing in a, so the unique optimum on $[0, d]$ is $\boxed{a^* = d}$. The conclusion assumes the stake is capped at the bankroll; otherwise positive expected profit could lead to unbounded borrowing in this idealized model.

Linear utility represents risk neutrality: an additional dollar has the same value regardless of current wealth. It can approximate preferences for small stakes, but is usually implausible for an entire bankroll. It ignores diminishing marginal utility and treats a 15% chance of losing all wealth as acceptable when expected wealth rises.

(b) Logarithmic utility

For $0 \le a < d$, define

$$g(a) = p \log(d + ba) + (1 - p) \log(d - a).$$

Its derivatives are

$$g'(a) = \frac{pb}{d + ba} - \frac{1 - p}{d - a}, \qquad g''(a) = -\frac{pb^2}{(d + ba)^2} - \frac{1 - p}{(d - a)^2} < 0.$$

Setting $g'(a) = 0$ gives $pb(d - a) = (1 - p)(d + ba)$ and thus

$$\boxed{\frac{a^*}{d} = \frac{pb - (1 - p)}{b} = \frac{p - q}{1 - q} = \frac{0.85 - 0.60}{0.40} = 0.625.}$$

Strict concavity makes this the unique maximum; $g(a) \to -\infty$ as $a \uparrow d$. Under literal even-money odds, the same derivation gives $a^*/d = 2p - 1 = 0.70$; part (a) still gives $a^* = d$.

(c) A hypothetical personal response

No: I would not commit \$312,500 of a \$500,000 net worth to this one event. At the quoted odds, final wealth would be approximately \$708,333 after a win and \$187,500 after a loss. The calculation is optimal under the stated probability, utility, and liquidity assumptions, but a probability estimate can be wrong, net worth need not be liquid, and essential future spending may justify greater risk aversion. Declining the full stake is consistent with changing these assumptions; it does not invalidate part (b). This is a hypothetical judgment, not a claim about an actual portfolio. With literal even-money odds, the proposed stake would instead be \$350,000.

2. Binary cross-entropy is a proper scoring rule

Write $\eta(x) = P(Y = 1 \mid X = x)$ and let $f(x) \in [0, 1]$ denote a predicted probability. Binary cross-entropy is

$$L(y, f(x)) = -y \log f(x) - (1 - y) \log(1 - f(x)).$$

Use the standard extended-value convention: a zero-probability term with zero weight contributes zero, and an impossible positive-weight event has infinite loss. Conditioning on $X = x$ gives

$$r_x(u) = \mathbb{E}[L(Y, u) \mid X = x] = -\eta(x) \log u - (1 - \eta(x)) \log(1 - u).$$

For $0 < \eta(x) < 1$ and $0 < u < 1$,

$$r'_x(u) = -\frac{\eta(x)}{u} + \frac{1 - \eta(x)}{1 - u} = \frac{u - \eta(x)}{u(1 - u)}, \qquad r''_x(u) = \frac{\eta(x)}{u^2} + \frac{1 - \eta(x)}{(1 - u)^2} > 0.$$

The unique conditional minimizer is $u = \eta(x)$. At the boundaries, $\eta(x) = 0$ gives $r_x(u) = -\log(1 - u)$, minimized at $u = 0$; $\eta(x) = 1$ gives $r_x(u) = -\log u$, minimized at $u = 1$.

By the tower property, global risk is $R(f) = \mathbb{E}_X[r_X(f(X))]$. More explicitly,

$$\begin{aligned} R(f) - R(\eta) &= \mathbb{E}_X\left[\eta(X) \log \frac{\eta(X)}{f(X)} + (1 - \eta(X)) \log \frac{1 - \eta(X)}{1 - f(X)}\right] \\ &= \mathbb{E}_X[D_{\mathrm{KL}}(\mathrm{Bern}(\eta(X)) \parallel \mathrm{Bern}(f(X)))] \geq 0. \end{aligned}$$

Equality holds precisely when $f(X) = \eta(X)$ almost surely. Thus binary cross-entropy is *strictly proper*, with probability vector $f^*(x) = (1 - \eta(x), \eta(x))$ if both class probabilities are reported.

This statement optimizes over all measurable probability predictors. A restricted linear-logit model need not contain the true conditional probability. Also, a finite sigmoid logit cannot attain 0 or 1, so in deterministic conditional cases the optimum over finite logits is an infimum rather than an attained minimum.

3. Why a constant SGD step size can be reasonable

The condition $\sum_t \rho_t^2 < \infty$ concerns asymptotically removing persistent gradient noise as $t \to \infty$, alongside other assumptions. A finite dataset and a finite training budget have a different goal: useful prediction, not arbitrarily precise minimization of the empirical objective.

For a local scalar quadratic with curvature $h > 0$, let the gradient noise ξ_t have mean zero and variance σ_g^2, independent across updates and of the current error. Constant-step SGD obeys

$$e_{t+1} = (1 - \rho h) e_t - \rho \xi_t.$$

When $0 < \rho h < 2$, the initial error contracts, but the limiting error variance is

$$\lim_{t \to \infty} \mathbb{E}[e_t^2] = \frac{\rho^2 \sigma_g^2}{1 - (1 - \rho h)^2} = \frac{\rho \sigma_g^2}{2h - \rho h^2} = O(\rho).$$

Thus a suitably small constant rate reaches a neighborhood of the optimum. If this optimization error is smaller than the statistical error from limited data, eliminating it further may yield little predictive benefit. A constant rate also avoids slowing too early; validation-based early stopping can control overfitting. Larger minibatches or iterate averaging reduce the noise further. This is not a general guarantee of exact convergence with a constant rate: that requires additional structure, such as vanishing gradient noise at an interpolating solution, or a later reduction in the rate.

4. Residual normality on the ALE dataset

Data and experimental design

The UCI ALE dataset contains 107 simulated sensor-network observations. Inputs are `anchor_ratio`, `trans_range`, `node_density`, and `iterations`; the target is `ale`, in meters. The column `sd_ale` is excluded. The original CSV is included with attribution and a SHA-256 integrity check [1].

I use `train_test_split(test_size=0.2, random_state=42)`: 85 training rows and 22 held-out rows. Models are fitted only to the training rows; StandardScaler statistics for OLS and SVR are learned there. All settings below were fixed before viewing test results. No model is selected by normality p-value. The same ALE split is reused in Problem 5.

Model	Settings
OLS	Standardized inputs; intercept included; sklearn LinearRegression.
RBF SVR	Standardized inputs; $C = 1$, $\epsilon = 0.1$, gamma=scale.
Decision tree	Maximum depth 3; minimum 5 samples per leaf; seed 42.
Random forest	300 trees; minimum 3 samples per leaf; seed 42.

For each model, residuals are $r_j = y_j - \widehat{f}(x_j)$ on the 22 test rows. I test the null that these residuals follow a normal distribution (with unspecified mean and variance), using SciPy's D'Agostino–Pearson omnibus statistic K^2, which combines skewness and kurtosis [3]. The nominal level is 0.05. I also report Holm-adjusted p-values across the four omnibus tests, and Shapiro–Wilk as a sensitivity check.

Results

Model	MAE (m)	K^2	Raw p	Holm p	Shapiro p
OLS	0.1954	7.300	0.0260	0.0780	0.2043
RBF SVR	0.2218	12.338	0.0021	0.0084	0.0089
Decision tree	0.1881	1.814	0.4037	0.8074	0.4093
Random forest	0.1976	1.551	0.4604	0.8074	0.1857

At the unadjusted 5% level, OLS and RBF SVR reject normality; the decision tree and random forest do not. After Holm correction, **only RBF SVR rejects**. The Shapiro–Wilk test also rejects for SVR, but not OLS, illustrating the limited stability of a small-sample normality conclusion. Q–Q plots on the next page show a prominent upper-tail departure for SVR and a large positive OLS residual.

Interpretation

These data do not support a blanket assertion that every fitted model has Gaussian residuals. Non-rejection for a model means insufficient evidence against normality, not proof of normality: 22 observations give limited power. Held-out residuals avoid the direct fitting constraints and possible interpolation of training residuals. Nevertheless, the test is a diagnostic under an iid sampling idealization; simulation design, systematic prediction bias, or changing variance across covariates can affect the interpretation. Marginal normality would not establish $Y \mid X \sim N(f(X), \sigma^2)$ or constant conditional variance.

Finally, normality is *not required* for squared error to be a sensible prediction loss. For square-integrable Y,

$$\mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(\mathbb{E}[Y \mid X] - f(X))^2] + \mathbb{E}[\mathrm{Var}(Y \mid X)].$$

Its unrestricted minimizer remains the conditional mean even with nonnormal errors.

4. Diagnostic plots

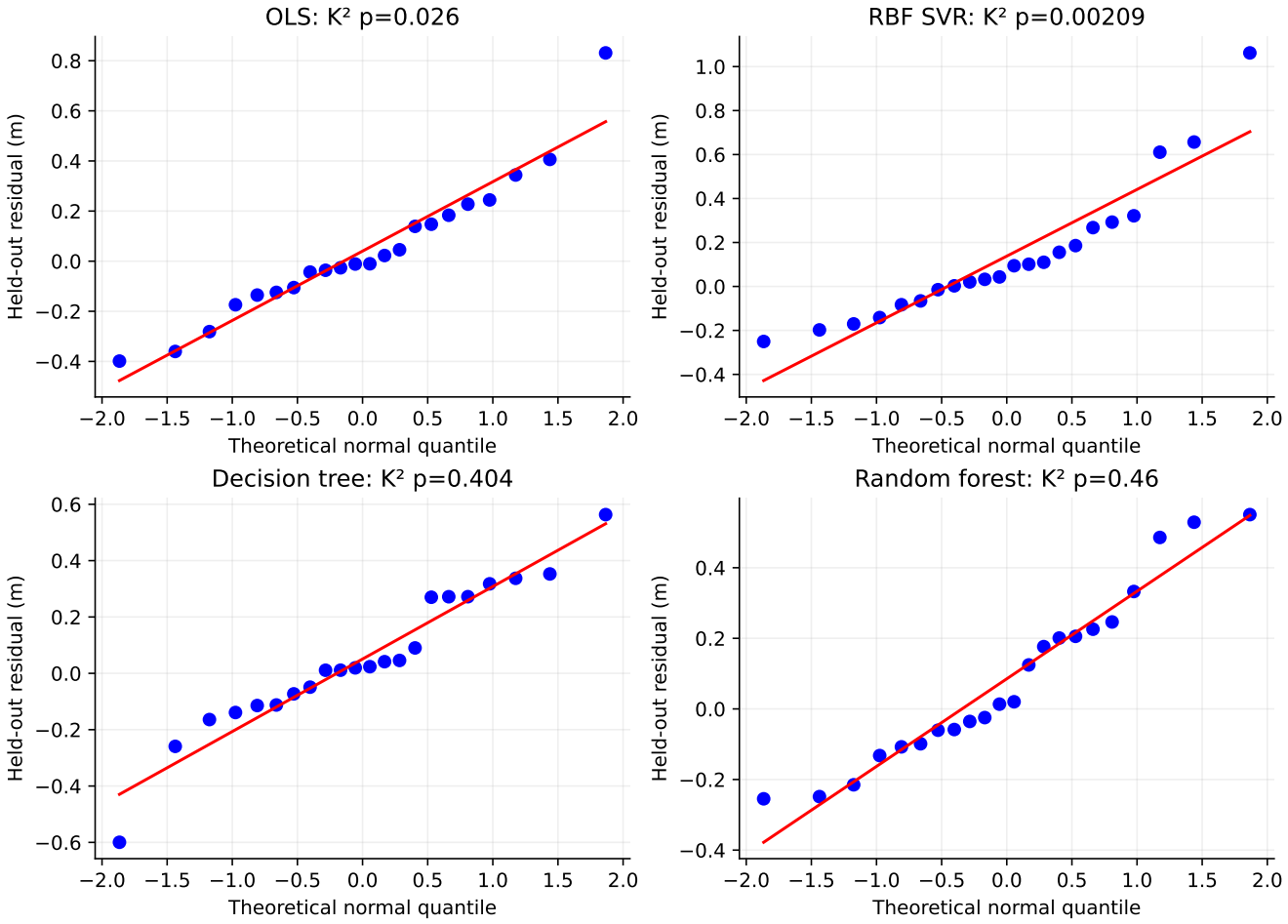


Figure 1. Normal Q–Q plots of held-out ALE residuals. The reference line is fitted by SciPy's `probplot`; straight-line behavior is expected for a normal distribution after allowing location and scale. Titles show unadjusted omnibus p-values, not the Holm-adjusted values.

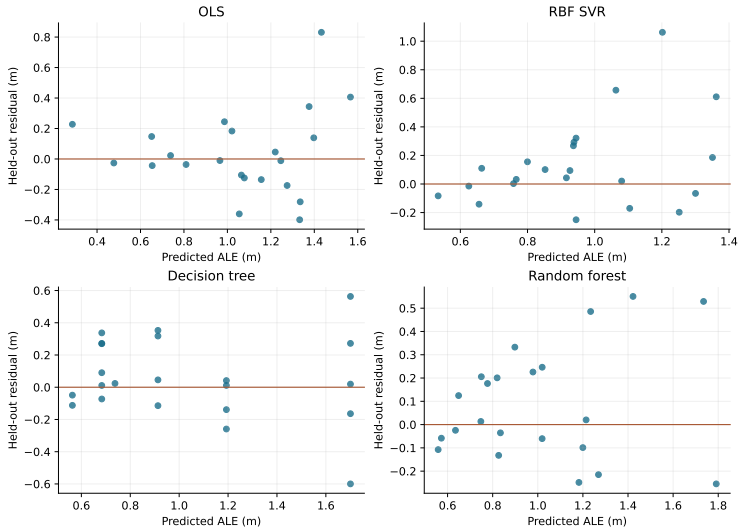


Figure 2. Residuals versus predicted ALE on the same test observations. Departures can reflect model mismatch as well as error-distribution shape.

5. Automatic differentiation: classification

Common setup and sklearn baseline

I load Breast Cancer Wisconsin (Diagnostic) using sklearn's `load_breast_cancer`: 569 observations and 30 features [2]. Labels are 0=malignant and 1=benign. The 80/20 split is stratified, with seed 42, giving 455 training and 114 test observations. Inputs are standardized using training means and standard deviations. The baseline is sklearn `LogisticRegression(C=1, solver="lbfgs", max_iter=10000, tol=1e-10)`. It retains default L2 regularization [5].

(a) Logistic model and binary cross-entropy

The PyTorch affine model computes $z_i = w^T \widetilde{x}_i + b$ and $p_i = \sigma(z_i)$. I implement the requested mean BCE as

$$\mathcal{L}_{\text{logit}}(w, b) = \frac{1}{n} \sum_i \big[\operatorname{softplus}(z_i) - y_i z_i \big], \qquad \operatorname{softplus}(z) = \log(1 + e^z).$$

This is algebraically identical to $-y \log p - (1-y) \log(1-p)$. PyTorch's stable softplus avoids overflow and cancellation; probabilities do not need to be clipped. Gradients are obtained with `loss.backward()`.

(b) Probit model with the same BCE

For probit, $p_i = \Phi(z_i)$. Since $1 - \Phi(z) = \Phi(-z)$, the same BCE is

$$\mathcal{L}_{\text{probit}}(w, b) = -\frac{1}{n} \sum_i \big[y_i \log \Phi(z_i) + (1 - y_i) \log \Phi(-z_i) \big].$$

I use `torch.special.log_ndtr` for stable log-CDF values [6] and `torch.special.ndtr` for probabilities. This avoids losing gradients through hard probability clipping in the tails.

Training and comparison

Both custom models use float64 CPU tensors, zero initial weights and bias, minibatches of 32, random reshuffling, and `torch.optim.SGD` with no momentum or weight decay. In epoch $e = 0, 1, \ldots$, the learning rate is $0.05/(1 + e/500)$. A further stratified split of the training set (364 inner training, 91 validation; seed 43) selects the minimum validation BCE among checkpoints every 10 epochs, up to 2,000. Scaling is refitted on the inner training subset for this step. Each final model is then trained from scratch on all 455 training examples for its selected epoch count. Test data are never used for this choice. Predictions use $p \geq 0.5$ for class 1.

Model	Correct / test	Accuracy	SGD epochs
sklearn logistic (L2)	112/114	98.25%	–
PyTorch logistic	111/114	97.37%	80
PyTorch probit	111/114	97.37%	20

Each custom classifier makes one more test error than the baseline, a difference of about 0.88 percentage points. The two custom links tie in accuracy on this split. This is not evidence of a general performance ranking: the test set is small, the baseline uses explicit L2 regularization, and the custom models use unpenalized BCE with early stopping. The comparison therefore includes these training differences as well as the link function. Saved confusion matrices and predictions permit a more detailed inspection.

5. Automatic differentiation: regression

Baseline and (a) squared-error model

Using the same 85/22 ALE split as Problem 4, the baseline is sklearn LinearRegression with an intercept. The PyTorch model is $\widehat{y}_i = w^T \widetilde{x}_i + b$, trained by minimizing

$$\mathcal{L}_{\text{MSE}}(w, b) = \frac{1}{n} \sum_i (y_i - w^T \widetilde{x}_i - b)^2.$$

This uses the lecture's convention without a factor 1/2. Only inputs are standardized; targets remain in their original units.

(b) Student-t residual likelihood

For $r_i = y_i - \widehat{y}_i$ and the assigned $\nu = 5$, unit-scale density,

$$p(r_i) = \frac{\Gamma(3)}{\sqrt{5\pi}\,\Gamma(5/2)} \left(1 + \frac{r_i^2}{5}\right)^{-3},$$
$$\mathcal{L}_t(w, b) = \frac{1}{n} \sum_i \left[\tfrac{1}{2}\log(5\pi) + \log\Gamma(5/2) - \log\Gamma(3) + 3\log\left(1 + \frac{r_i^2}{5}\right)\right].$$

The implementation uses the negative mean of `StudentT(df=5, loc=0, scale=1).log_prob(residual)` [7]. The degrees of freedom and scale are fixed, not fitted. In particular, rescaling y before applying a unit-scale density would change the assigned loss, so I do not do that.

For a single example,

$$\frac{\partial L_t}{\partial \widehat{y}_i} = -\frac{6 r_i}{5 + r_i^2}, \qquad \frac{\partial L_{\text{MSE}}}{\partial \widehat{y}_i} = -2 r_i.$$

The Student-t gradient downweights large residuals, while the MSE gradient grows linearly. This explains robustness to large errors, but does not guarantee a lower MAE on every test set. The Student-t objective is not globally convex in the residual for $|r| > \sqrt{5}$.

Training, results, and optimization checks

Both custom regressions use zero initial weights, bias initialized to the training mean of ALE, 5,000 epochs, minibatches of 16, and ordinary SGD with rate $0.02/(1 + e/100)$. Settings are fixed without test tuning. Primary evaluation is test mean absolute error, $n_{\text{test}}^{-1} \sum_j |y_j - \widehat{y}_j|$.

Model	Test MAE (m)	Test RMSE (m)	Train MSE
sklearn OLS	0.195393	0.270543	0.050582
PyTorch linear, MSE	0.195373	0.270370	0.050582
PyTorch linear, Student-t	0.194071	0.272702	0.050627

The custom MSE fit nearly reproduces OLS: its training MSE is only 3.35×10^{-7} above the OLS optimum, and the largest test-prediction difference is 0.001148 meters. Small test-MAE differences here are optimization error, not a change in model class or loss.

Student-t regression lowers test MAE by 0.001322 meters (0.68%) relative to OLS on this split, while its RMSE is slightly higher. Thus it trades a small improvement in average absolute error for worse squared-error performance here. Its training negative log-likelihood is 9.76×10^{-8} above an independent BFGS reference solution. That reference only checks optimization; all reported model predictions come from SGD. The small MAE improvement on 22 test observations should not be presented as established statistical superiority.

Optimization diagnostics and reproducibility

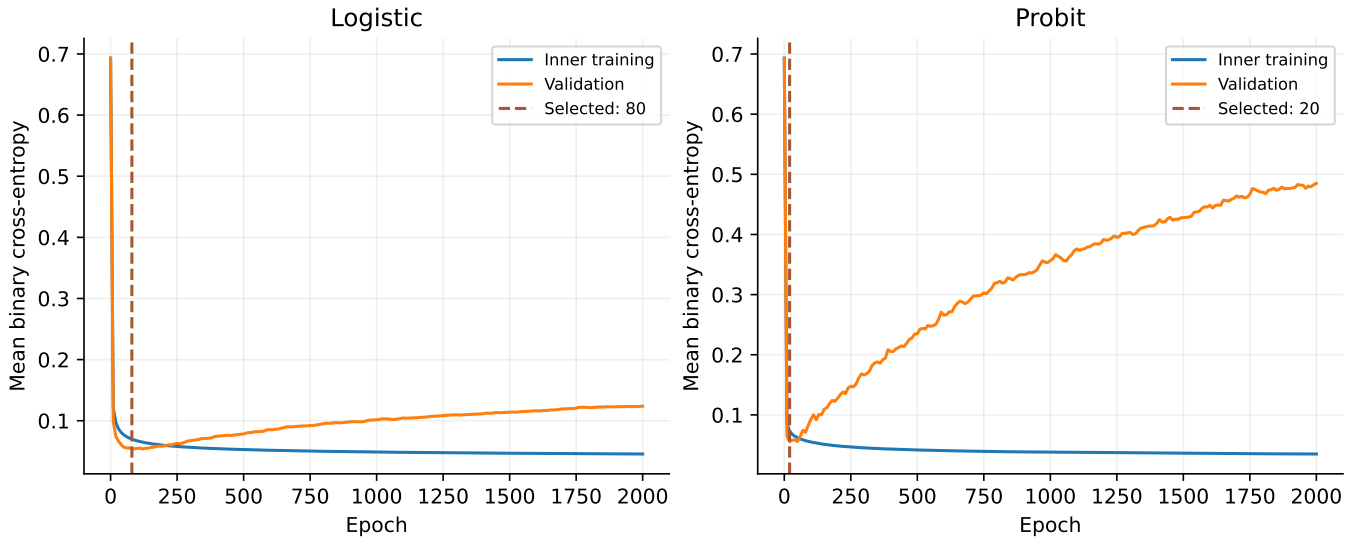


Figure 3. Training-only classification epoch selection. Vertical lines mark the chosen epochs before refitting on the entire training set.

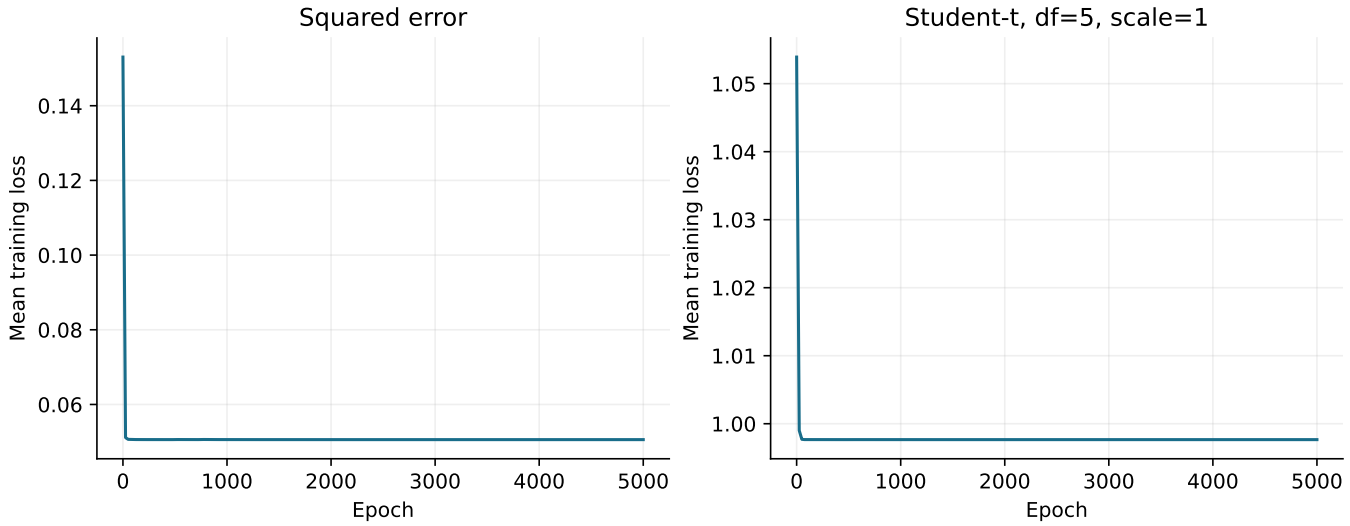


Figure 4. Mean training objectives for the two ALE regressions. The vertical scales differ because the losses have different definitions.

Reproduction. The repository stores code under `homework_1/`. Create a Python 3.12 environment, install `requirements.txt`, and execute `python homework_1/run.py`. This regenerates every empirical table, prediction file, and figure. Then run `python -m unittest discover -s homework_1` with `-p 'test_*.py' -v` for eight checks of the custom losses, analytic and finite-difference gradients, tail stability, data integrity, and recovery of a known linear solution. `python homework_1/build_report.py` refreshes report tables and compiles this PDF if `pdflatex` is installed.

Audit trail. JSON files record fixed seeds, train/test and validation indices, original and standardized coefficients, preprocessing statistics, package versions, source hashes, and the data checksum. CSV files expose per-observation predictions and optimization histories. Runs use one CPU thread and float64. Dependencies are pinned; tiny floating-point differences across platforms remain possible. No external credentials are required to run the code.

References

1. Singh, A., Kotiyal, V., Sharma, S., Nagar, J., and Lee, C. (2020). *Average Localization Error (ALE) in sensor node localization process in WSNs*. UCI Machine Learning Repository. doi:10.24432/C5ZP6R. The public CSV is redistributed unchanged under CC BY 4.0, with its filename shortened to `ale.csv`.
2. Wolberg, W., Mangasarian, O., Street, N., and Street, W. (1993). *Breast Cancer Wisconsin (Diagnostic)*. UCI Machine Learning Repository. doi:10.24432/C5DW2B. Loaded through sklearn's bundled dataset.
3. SciPy. `scipy.stats.normaltest`: D'Agostino–Pearson omnibus normality test.
4. SciPy. `scipy.stats.probplot`: probability-plot construction and fitted reference line.
5. scikit-learn. `LogisticRegression`: default regularization and solver parameters.
6. PyTorch. `torch.special`: normal CDF and stable log-CDF functions.
7. PyTorch. `torch.distributions.StudentT`: Student-t log probability with specified degrees of freedom, location, and scale.
8. PyTorch. `torch.optim.SGD`: stochastic gradient descent optimizer.
9. Veitch, V. Course materials supplied with this assignment: *Lecture 1*; *Supervised Learning Continued*; *First Look at Generalization*; *Feature Engineering and Deep Learning*; and *Homework 1*, Fall 2026. The theoretical arguments use the notation and risk-minimization framework of these lectures.